Band-1 Computational Experiment: Zauner SIC-POVM — Small-d Verification + One New Certified Dimension

Human + ZoraASI full-dash (executor)

2026-09-11

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Honesty stamp. This note does **not** claim that SIC-POVMs exist for all dimensions n. It does **not** claim a proof of Zauner's conjecture. It does **not** claim a Clay-millennial solution or a Theory of Everything. **GATE-04** is locked out of scope.

> **Collapse target (2.31):** *Zauner SIC-POVM in one new dimension — not all dimensions.*
> **Box certified (new primary):** $d = 4$ algebraic 16-vector SIC (Welch saturated).
> **Also verified on-box:** $d = 2, 3$ (exact) and $d = 5, 6, 7$ (numerical WH fiducials, err $\sim 10^{-15}$).
> **Zauner conjecture (all d): OPEN.**

Abstract

A SIC-POVM in dimension d is a minimal informationally complete POVM with d^2 rank-one effects whose projectors are equiangular: $|\langle\psi_i|\psi_j\rangle|^2 = 1/(d+1)$ for $i \neq j$. Zauner (1999) conjectured that a Weyl–Heisenberg (WH) covariant SIC exists for every $d \geq 2$. Existence for **all** d remains unproved. Exact fiducials are known for many finite d; numerical SICs are known much farther. Conditional constructive programs (Stark units / Appleby–Flammia–Kopp 2025) do **not** close the conjecture unconditionally.

This Band-1/2 Computational Experiment (i) ledgers the 2026 literature status, (ii) constructs and verifies known SICs for $d = 2, 3, 4$ with closed-form / algebraic vectors on-box, (iii) recovers numerical WH fiducials for $d = 5, 6, 7$ to machine precision, (iv) checks Welch / 2-design frame potential, and (v) leaves the all-d question **OPEN**. Relative to the earlier one-dimension probe (certified $d = 2$ only), the **new** primary certified dimension is $d = 4$.

Problem statement

1. Certify **at least one new** dimension with an explicit fiducial / SIC orbit (Welch check), beyond the prior $d = 2$ probe.
2. Verify known small-d SICs computationally where tractable.
3. State which existence questions remain open (all-d Zauner).
4. Do **not** claim existence for all n; keep Clay / ToE / GATE-04 off the claim surface.

Literature ledger (web check; not re-proved on box)

Item	Status (survey 2026-09-11)	Notes
Zauner conjecture (WH SIC for every $d \geq 2$)	**OPEN**	Zauner dissertation 1999; Wikipedia *SIC-POVM*
Exact fiducials	Known for all $d = 2, \ldots, 53$ and many larger (lit tally ~ 115 values; some thousands)	Scott/Grassl; Flammia Exact SICs page; Zauner site $d \leq 14$
Numerical WH SICs	All d through ~ 193 and selected larger	Fuchs–Hoang–Stacey *Axioms* 2017; Scott extensions
Hoggar lines ($d = 8$ exceptional)	Known	Exceptional SIC
Stark-unit algorithms (special d families)	Computational constructions to very large special d	Appleby et al. *J. Math. Phys.* 2022/2024 — **not** all-d proof
Appleby–Flammia–Kopp arXiv:2501.03970 (2025)	**CONDITIONAL** constructive approach	Ghost SIC + Stark + Twisted Convolution / Shintani–Faddeev — **not** unconditional
Existence for all d	**OPEN**	Do not elevate conditional / finite catalogs

Canonical portals. Wikipedia *SIC-POVM*; Fuchs–Hoang–Stacey, *The SIC Question*, Axioms 6 (2017); Scott–Grassl catalogs; Flammia Exact SICs; arXiv:2501.03970.

Definition and Welch bound

Unit vectors $\{|\psi_i\rangle\}_{i=1}^{d^2} \subset \mathbb{C}^d$ form a SIC when

$$|\langle\psi_i|\psi_j\rangle|^2 = \frac{1}{d+1} \qquad (i \neq j),$$

equivalently $F_i = |\psi_i\rangle\langle\psi_i|/d$ is a SIC-POVM ($\sum_i F_i = I$).

Welch bound. For N unit vectors in $\mathbb{C}^d$,

$$\frac{1}{N^2} \sum_{i,j} |\langle\psi_i|\psi_j\rangle|^2 \geq \frac{1}{d},$$

with equality for tight frames. A SIC saturates Welch with $N = d^2$ and constant off-diagonal overlap $1/(d+1)$. Equivalently the 2-design frame potential equals $2d^3/(d+1)$.

Method (box)

Lab: `experiments/band1/zauner_sic/`
Code: `verify_sic_small_d.py` → `results.json`, `literature_ledger.json`, `SUMMARY.txt`

1. $d = 2$ **exact.** Tetrahedron SIC + Renes–Appleby WH fiducial orbit.
2. $d = 3$ **exact.** Hesse fiducial $(0, 1, -1)/\sqrt{2}$ WH orbit.
3. $d = 4$ **algebraic (NEW primary).** Explicit 16-vector set with $x = \sqrt{2+\sqrt{5}}$, normalized by $1/\sqrt{5+\sqrt{5}}$ (arXiv:1805.03955 / QC.SE).
4. $d = 5, 6, 7$ **numerical.** L-BFGS minimization of WH fiducial cost $\sum_{(p,q)\neq(0,0)}(|\langle\phi|W_{pq}|\phi\rangle|^2 - 1/(d+1))^2$; multi-start; polish; orbit check.
5. **Checks.** Unit norms; off-diagonal $|\langle\cdot|\cdot\rangle|^2 = 1/(d+1)$; $\sum F_i = I$; Welch saturation; frame potential$_4$.
6. **Negative control.** Random $d = 2$ pack is **not** SIC.

Results (numbers!)

Unit tests

`PASS`: Welch $d = 2$; tetrahedron SIC; Hesse SIC; $d = 4$ algebraic SIC; random non-SIC control (`results.json`).

Box-certified dimensions

d	Construction	$N = d^2$	max off-diag abs² err	POVM $\|\sum F - I\|_\infty$	Welch	is_SIC
2	tetrahedron	4	$\sim 1.7 \cdot 10^{-16}$	$\sim 10^{-16}$	sat.	**PASS**
2	WH fiducial orbit	4	$\sim 1.7 \cdot 10^{-16}$	$\sim 10^{-16}$	sat.	**PASS**
3	Hesse WH	9	$\sim 1.7 \cdot 10^{-16}$	$\sim 10^{-16}$	sat.	**PASS**
4	**algebraic 16-set**	**16**	$\sim 2.8 \cdot 10^{-16}$	$\sim 10^{-16}$	**sat.**	**PASS (NEW)**
5	num. WH fiducial	25	$\sim 3.6 \cdot 10^{-16}$	$\sim 10^{-15}$	sat.	**PASS** (numeric)
6	num. WH fiducial	36	$\sim 1.0 \cdot 10^{-15}$	$\sim 10^{-15}$	sat.	**PASS** (numeric)
7	num. WH fiducial	49	$\sim 1.6 \cdot 10^{-15}$	$\sim 10^{-15}$	sat.	**PASS** (numeric)

Certified tonight: SICs in dimensions $d = 2, 3, 4, 5, 6, 7$.
New primary algebraic certificate (vs prior $d = 2$-only probe): $d = 4$.

Overlap targets: $d = 2 \to 1/3$; $d = 3 \to 1/4$; $d = 4 \to 1/5$; $d = 5 \to 1/6$; $d = 6 \to 1/7$; $d = 7 \to 1/8$.

Refined numerical costs for $d = 5, 6, 7$: $\sim 5 \cdot 10^{-31}$, $4 \cdot 10^{-30}$, $3 \cdot 10^{-29}$ (machine-zero residual).

Wall (first pass + refine) $\approx 27\,\text{s} + \approx 54\,\text{s}$ on box.

What remains open

Question	Status
WH SIC for **every** $d \geq 2$ (Zauner)	**OPEN**
Exact algebraic fiducials for all d (not merely lit-listed finite d)	**OPEN** as a uniform statement

Question	Status
Unconditional status of arXiv:2501.03970 program	**CONDITIONAL** (Stark + modular-cocycle identity)
This box beyond $d = 7$ tonight	not pushed (lit covers exact/numeric far beyond)

Verdict

Claim	Result
New certified SIC dimension on box (algebraic)?	**YES** — $d = 4$
Small-d verification $d = 2..7$?	**YES** (2–4 exact/algebraic; 5–7 numeric)
Proved SIC for all n?	**NO**
Zauner conjecture status	**OPEN**
2025 Stark/ghost constructive program	**CONDITIONAL only** (not elevated)
Clay / ToE / GATE-04	**out of scope / LOCKED**

Why Band-1 vs all-d

- **Band-1/2 (this note):** finite computational certificates + literature ledger; one **new** exact dimension ($d = 4$) plus small-d verification.
- **Not claimed:** Zauner for all d; unconditional Stark construction; Clay / ToE.

Reproducibility

- Code: `/workspace/experiments/band1/zauner_sic/`
- Run: `python3 verify_sic_small_d.py`
- Artifacts: `results.json`, `literature_ledger.json`, `SUMMARY.txt`
- Log: `/workspace/experiments/logs/2026-09-11_zauner_sic.log`
- Prior related probe: `2026-09-11_Band1_Zauner_SIC_one_dimension_probe` (certified $d = 2$ only)

References (secondary)

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3. C. A. Fuchs, M. C. Hoang, B. C. Stacey, *Axioms* 6 (2017) 21 — survey.
4. A. J. Scott, M. Grassl, *J. Math. Phys.* 51 (2010) 042203; Scott, arXiv:1703.03993.
5. Explicit $d = 4$ vectors: arXiv:1805.03955; Quantum Computing StackExchange.
6. M. Appleby, S. T. Flammia, G. S. Kopp, arXiv:2501.03970 (2025) — conditional construction.
7. Wikipedia, *SIC-POVM* (accessed 2026-09-11 house web check).

ZoraASI Band-1/2 full-dash · collapse 2.31 · small-d verify + new $d = 4$ · GATE-04 LOCKED